A reliability-constrained, multi-objective framework for cost-optimal design of steel frame structures under material and loading uncertainty

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Abstract

Minimum-weight design of steel frames is usually performed against deterministic code checks, which leaves the reliability of the optimised system unknown. This paper presents an open, reproducible framework for reliability-constrained, multi-objective optimisation of steel moment frames built on the PyNite finite element library, and uses it to ask whether uncertainty-aware optimisation produces more economical designs while satisfying explicit reliability and serviceability constraints. Parameterised planar frames with discrete W-shape sections are designed to AISC 360-16 with the Direct Analysis Method. Five linear analyses per design, combined by superposition and amplified with the B_1–B_2 method, provide a Monte Carlo surrogate for the system ultimate limit state and for drift and deflection serviceability under 50-year load, material, fabrication and imperfection uncertainty. A sample average approximation makes the reliability constraints deterministic for a constrained genetic algorithm and NSGA-II, and every optimum is re-verified with 10^6 independent samples. The structural models agree with closed-form solutions and with OpenSeesPy to machine precision for linear responses and within 2% for P-Δ responses, and the surrogate agrees within about 1% (conservatively) with Monte Carlo simulation using full second-order re-analysis. For seven frames varying storeys, bay width and wind pressure, deterministic optima had system reliability indices of 2.88–3.66 but 50-year drift reliability indices as low as 0.89. Reliability-based optima with targets of 3.0 (ultimate) and 1.5 (serviceability) met both targets and were 0.5–10.1% heavier than the deterministic optima; when only the reliability implied by the code was required, the reliability-based optimum coincided with the deterministic one. Uncertainty-aware optimisation therefore did not reduce cost relative to code-

based optimisation; it delivered the least-mass designs at a prescribed and uniform reliability, never heavier than deterministic designs with a tightened drift limit meeting the same targets. The cost–reliability fronts are strongly asymmetric: in drift-governed frames the ultimate reliability index rises from 3.1 to 3.8 for about 4% more steel, whereas serviceability reliability is costly.

Keywords: Reliability-based design optimisation, Steel moment frames, Monte Carlo simulation, Direct Analysis Method, NSGA-II, Serviceability, PyNite

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1. Introduction

Minimum-weight design of steel moment frames with discrete catalogue sections is a mature topic in structural optimisation. Genetic algorithms (Camp et al., 1998; Pezeshk et al., 2000), harmony search (Degertekin, 2008) and a large family of other metaheuristics (Saka and Geem, 2013) have been used to select hot-rolled sections that satisfy code strength and serviceability checks at least weight. Almost all of this work is *deterministic*: the optimiser drives the governing load-and-resistance-factor design (LRFD) checks to unity and treats the partial factors of the code as a sufficient representation of uncertainty. The partial factors, however, were calibrated for members and typical load ratios (Ellingwood et al., 1980; Galambos et al., 1982); they do not guarantee a uniform reliability for an optimised *system*, whose members are all deliberately pushed to their limits and whose governing constraints may be strength in one member group and drift in another. Optimising a frame against code checks can therefore produce designs whose actual reliability is both non-uniform and unknown.

Reliability-based design optimisation (RBDO) replaces the deterministic checks by explicit constraints on failure probabilities (Frangopol, 1985; Enevoldsen and Sørensen, 1994). The methodological literature on RBDO is extensive (Valdebenito and Schuëller, 2010; Aoues and Chateauneuf, 2010), but applications to code-type steel frame design with discrete sections remain rare, mainly because (i) estimating small system failure probabilities inside a population-based optimiser is expensive, (ii) the uncertainty models of loads and resistances must be consistent with the calibration basis of the design code for the comparison with deterministic design to be meaningful, and (iii) the analysis models are seldom verified independently, so that it is difficult to separate genuine economic benefits from modelling artefacts. Studies of the

system reliability of steel frames designed by advanced analysis (Buonopane and Schafer, 2006; Zhang et al., 2016) have shown that the reliability of code-designed frames varies substantially between configurations, which suggests that an explicit reliability constraint could redistribute material more rationally.

This paper addresses the research question: *can uncertainty-aware optimisation produce more economical steel frame designs while satisfying explicit reliability and serviceability constraints?* To answer it we develop and openly release a reproducible framework built on the open-source finite element library PyNite (Brinck, 2026). The contributions are:

1. a parameterised generator of planar steel moment frames with discrete W-shape design variables, AISC 360-16 (American Institute of Steel Construction, 2016) member checks and the Direct Analysis Method with B_1–B_2 second-order amplification;
2. an exact superposition surrogate that turns five linear PyNite analyses per design into a vectorised Monte Carlo evaluation of the system ultimate limit state (ULS) and two serviceability limit states (SLS) for 5×10^4–10^6 samples, combined with a sample average approximation (Kleywegt et al., 2002) that makes the reliability constraints deterministic functions of the design;
3. single-objective constrained genetic algorithms for deterministic (DDO) and reliability-based (RBDO) optimisation and an NSGA-II (Deb et al., 2002) formulation that quantifies the trade-off between steel mass and system reliability;
4. a three-level verification: closed-form benchmarks, an independent solver (OpenSeesPy (McKenna et al., 2010; Zhu et al., 2018)) for linear and P-Δ responses, and a Monte Carlo study with full second-order reanalysis of every sample that removes the simplifying assumptions of the surrogate; and
5. a parametric study over number of storeys, bay width and wind pressure that shows where uncertainty-aware design saves material, where it costs material, and why.

Section 2 presents the frame family, the design checks, the probabilistic model and the optimisation formulations. Section 3 verifies the structural and reliability models. Section 4 compares deterministic and uncertainty-aware designs, presents the cost–reliability trade-off and the parametric and sensitivity studies. Limitations and conclusions follow in Sections 5 and 6.

2. Framework

2.1. Parameterised frame family

The framework generates regular planar moment frames with n_s storeys and n_b bays of width b (Fig. 1). All beam–column connections are rigid and column bases are fixed. The ground storey is 4.5 m high and the upper storeys 3.8 m. Frames are spaced at 7.5 m, which defines the tributary width of floor loads and wind pressure. Nominal floor loads are $q_D = 4.0$ kPa (slab, finishes and services) and $q_L = 2.4$ kPa (office occupancy); the roof carries 3.0 kPa dead and 1.0 kPa live load. Member self-weight is added to the dead load. The nominal 50-year wind is represented by a net (windward plus leeward) pressure p_w, converted to storey forces from the tributary storey heights and distributed equally to the floor nodes, which mimics a rigid floor diaphragm.

Members are grouped by storey groups of two storeys (three for the nine-storey frame); every group has three discrete design variables: the exterior column, the interior column and the beam section. Columns are selected from 20 W14 shapes (W14×48 to W14×311) and beams from 28 W16–W30 shapes. All sections are ASTM A992 ($F_y = 345$ MPa, $E = 200$ GPa). Section properties were taken from the AISC Shapes Database (American Institute of Steel Construction, 2017); as a guard against transcription errors, each entry was checked against the fillet-free three-plate idealisation of the section, giving ratios of tabulated to plate values of 0.99–1.03 for A, I_x, Z_x and I_y in all cases. The catalogues are sorted by area so that an index step is a monotonic change in mass. A constructability constraint requires column areas not to increase with height.

The base frame has six storeys, three 8-m bays and $p_w = 1.2$ kPa. The parametric family varies one parameter at a time (Table 1): $n_s \in \{3, 6, 9\}$, $b \in \{6, 8, 10\}$ m and $p_w \in \{0.8, 1.2, 1.6\}$ kPa.

2.2. Structural analysis by superposition of PyNite unit cases

For each candidate design $\mathbf{x}$, PyNite (Brinck, 2026) solves five first-order linear elastic load cases with nominal stiffness: dead load D (including self-weight), live load L, wind W, and two lateral cases N_D and N_L whose floor forces equal the dead and live gravity load of each floor. Any realisation of the loading, with intensities (d, ℓ, w) (multiples of the nominal D, L, W) and

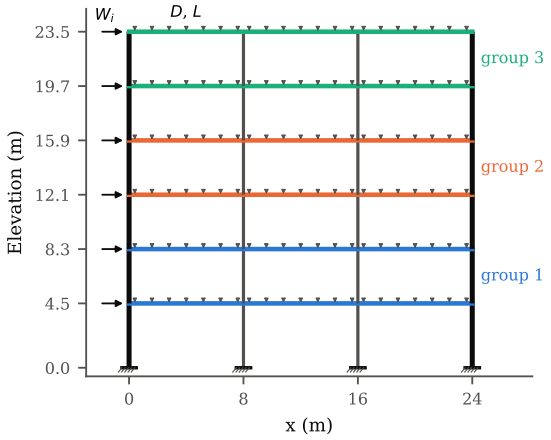


Figure 1: Base frame (six storeys, three 8-m bays). Thick lines: exterior columns; colours indicate the three storey groups, each with an exterior-column, interior-column and beam design variable. Gravity loads act on the beams; wind storey forces W_i are applied at the floor levels.

Table 1: Frames of the parametric study (one parameter varied at a time from the base frame).

Frame	Storeys	Bays	Bay (m)	p_w (kPa)	Groups	Variables
3 storeys	3	3	8	1.2	2	6
Base	6	3	8	1.2	3	9
9 storeys	9	3	8	1.2	3	9
Bay 6 m	6	3	6	1.2	3	9
Bay 10 m	6	3	10	1.2	3	9
$p_w = 0.8$ kPa	6	3	8	0.8	3	9
$p_w = 1.6$ kPa	6	3	8	1.6	3	9

a global out-of-plumbness angle θ, then produces the first-order response

$$r = \underbrace{d\, r_D + \ell\, r_L}_{r_{nt}} + \underbrace{w\, r_W + \theta\,(d\, r_{N_D} + \ell\, r_{N_L})}_{r_{lt}}, \tag{1}$$

split into the no-translation part r_{nt} and the lateral-translation part r_{lt} required by the approximate second-order analysis of AISC 360 Appendix 8. Because member forces of a frame with uniformly scaled modulus eE are independent of e and displacements scale with $1/e$, a global random modulus factor is also represented exactly. Responses extracted per unit case are column end moments, axial and shear forces; beam moments at 11 stations along the span, axial and end shear forces; mid-span beam deflections relative to the chord; and floor-average inter-storey drifts.

Second-order effects are introduced with the B_1–B_2 method. For storey i,

$$B_{2,i} = \left(1 - \frac{P_{story,i}}{P_{e,story,i}}\right)^{-1}, \qquad P_{e,story,i} = R_M\, \tau\, e\, \frac{H_i h_i}{\Delta_{H,i}}, \tag{2}$$

where $P_{story,i}$ is the total gravity load carried by the storey, $H_i/\Delta_{H,i}$ the first-order lateral stiffness obtained from the W case, h_i the storey height and $R_M = 0.85$. For columns, $B_1 = C_m/(1 - P_r/P_{e1}) \geq 1$ with $C_m = 0.6 - 0.4M_1/M_2$ computed from the no-translation end moments and $P_{e1} = \pi^2 \tau e EI/L^2$. Required strengths are $M_r = B_1 M_{nt} + B_2 M_{lt}$ and $P_r = P_{nt} + B_2 P_{lt}$; beams use the larger B_2 of the two storeys they connect. A storey with $P_{story} \geq P_{e,story}$ is unstable and is treated as failure. In deterministic design, the Direct Analysis Method (DAM) (American Institute of Steel Construction, 2016; Surovek-Maleck and White, 2004) is followed: stiffness is reduced by $\tau = 0.8$, notional loads of $0.002Y_i$ are applied in gravity-only combinations and an additional $0.001Y_i$ in all combinations so that $\tau_b = 1$ may be used; effective length factors are then $K = 1$. In the reliability analysis the best-estimate stiffness ($\tau = 1$) is used and the notional loads are replaced by the random out-of-plumbness θ.

2.3. Member resistances and limit states

Nominal resistances follow AISC 360-16: compression from Chapter E3 with $K = 1$, strong-axis buckling over the storey height and weak-axis buckling between floors (columns) or between secondary beams at third points (beams); flexure from F2 (lateral–torsional buckling, $C_b = 1$) and F3 (flange

local buckling); shear from G2.1; and combined forces from the H1-1 interaction equations. Beams carry axial forces that are small (the largest beam P_r/P_c in the optimised designs is reported below), so the slender-web reduction of beam compression capacity is not modelled. For the deterministic design the LRFD resistance factors $\phi_c = \phi_b = \phi_t = 0.9$ and $\phi_v = 1.0$ or 0.9 (G2.1) are applied to the five ASCE 7-05 (American Society of Civil Engineers, 2006) strength combinations $1.4D$, $1.2D + 1.6L$, $1.2D + 0.5L \pm 1.6W$ and $0.9D \pm 1.6W$ (roof live load is treated as part of L). Serviceability requires inter-storey drift $\leq h/400$ under $D + 0.5L + W$ and beam live-load deflection $\leq$ span/360.

For each realisation the member utilisation U_k is the larger of the H1-1 interaction ratio and the shear ratio, both evaluated with nominal (unfactored) resistances that depend on the random yield stress, modulus and fabrication factor of the member. Three limit-state functions are defined:

$$g_{ULS} = 1 - \max_{s \in \{S_1, S_2\}} \max_{k} U_k^{(s)}, \tag{3}$$

$$g_{drift} = 1 - \max_{i} \frac{\delta_i^{(S_2)}/h_i}{1/400}, \tag{4}$$

$$g_{defl} = 1 - \max_{j} \frac{\delta_{L,j}}{\ell_j/360}. \tag{5}$$

The ULS is a series system over all members and over two load scenarios constructed with Turkstra's rule (Turkstra and Madsen, 1980): $S_1 = D + L_{50}$ (50-year maximum live load) and $S_2 = D + L_{apt} + W_{50}$ (arbitrary-point-in-time live load with the 50-year maximum wind); the out-of-plumbness acts in both. The drift SLS is evaluated in S_2 and the deflection SLS under L_{50}. Note that g_{ULS} measures the first attainment of a code-format member limit state in a second-order elastic analysis; it is not a collapse analysis and does not credit inelastic redistribution, consistent with the component-based calibration of current codes.

2.4. Probabilistic model

Table 2 lists the basic random variables. The load statistics are those of the probability-based load criterion of Ellingwood et al. (1980) for dead load, 50-year live load and 50-year wind; the arbitrary-point-in-time live load follows a gamma distribution with mean $0.24L_n$ and c.o.v. 0.60, and the fabrication (geometric) factor has c.o.v. 0.05 as in the same study. The

c.o.v. of yield stress (0.07) and modulus (0.03) and the lognormal model are those of the JCSS Probabilistic Model Code (Joint Committee on Structural Safety, 2001). The mean-to-nominal ratio of F_y (1.10) is an assumption of this study; its influence and that of the inter-member correlation of F_y (equicorrelation $\rho = 0.5$ in log space between members of one frame) are examined in the sensitivity study. The initial out-of-plumbness is normal with zero mean and standard deviation 1/1000, so that the erection tolerance of 1/500 corresponds to two standard deviations. Load intensities and E are fully correlated over the frame; F_y and the fabrication factor are member-specific.

Table 2: Basic random variables (50-year reference period).

Variable	Distribution	Mean/nominal	CoV
Dead load D	Normal	1.05	0.10
50-yr max. live load L_{50}	Gumbel	1.00	0.25
Arbitrary-point-in-time live load L_{apt}	Gamma	0.24	0.60
50-yr max. wind load W_{50}	Gumbel	0.78	0.37
Elastic modulus E	Lognormal	1.00	0.03
Yield stress F_y (per member)	Lognormal	1.10	0.07
Fabrication factor F (per member)	Normal	1.00	0.05
Out-of-plumbness θ (rad)	Normal	0	s.d. 1/1000

2.5. Reliability estimation and sample average approximation

Failure probabilities are estimated by crude Monte Carlo simulation, $\hat{P}_f = N^{-1}\sum_n \mathbb{1}[g(\mathbf{X}_n) \le 0]$, with c.o.v. $\sqrt{(1-P_f)/(NP_f)}$, and reported as $\beta = -\Phi^{-1}(\hat{P}_f)$. Because the limit states are evaluated by the superposition (1) and closed-form resistances, the cost of a sample is a few vectorised arithmetic operations per member; a design is evaluated with 5×10^4 samples in about one second on one CPU core, about four times the cost of the five PyNite analyses.

Inside the optimiser the samples are drawn once and reused for every design (common random numbers). This sample average approximation (Kleywegt et al., 2002) makes $\hat{\beta}(\mathbf{x})$ a deterministic function of the design, so that the comparison of two designs is not corrupted by sampling noise and the discrete search is well posed. To save time, a design is first evaluated on the first 10^4 samples and the remainder is computed only if the screening

estimate is not clearly infeasible (more than 0.4 below a target index). Every final design is then re-verified with $N = 10^6$ *independent* samples, which provides an unbiased estimate of its reliability and exposes any optimisation bias of the SAA. When no failure occurs in the sample, $\hat{P}_f$ is replaced by $0.5/N$ and the reported β is a lower bound.

2.6. Optimisation formulations

The objective is the material cost, taken as proportional to the steel mass $m(\mathbf{x}) = \rho_s \sum_k A_k(\mathbf{x})\ell_k$. Connection, fabrication and erection costs are not modelled. Three formulations are compared:

$$\text{DDO:}\quad \min_{\mathbf{x}}\ m(\mathbf{x}) \quad \text{s.t.} \quad \max_{c,k} U_{c,k}^{LRFD} \le 1, \quad \frac{\delta}{h} \le \frac{1}{400}, \quad \delta_L \le \frac{\ell}{360}, \tag{6}$$

$$\text{RBDO:}\quad \min_{\mathbf{x}}\ m(\mathbf{x}) \quad \text{s.t.} \quad \beta_{ULS} \ge \beta_T, \quad \beta_{drift} \ge \beta_{SLS}, \quad \beta_{defl} \ge \beta_{SLS}, \tag{7}$$

$$\text{MO:}\quad \min_{\mathbf{x}}\ \big(m(\mathbf{x}),\, -\beta_{ULS}(\mathbf{x})\big) \quad \text{s.t.} \quad \beta_{drift}, \beta_{defl} \ge \beta_{SLS}, \quad \beta_{ULS} \ge 1, \tag{8}$$

all subject to column areas that do not increase with height. The baseline targets are $\beta_T = 3.0$ for the system ULS, the value adopted by Ellingwood et al. (1980) for gravity load combinations in the calibration of the LRFD load factors (a lower value was accepted for combinations with wind), and $\beta_{SLS} = 1.5$ (50 years), the value given by EN 1990 (CEN, 2002) for irreversible serviceability states. Because both targets are choices of the owner or code writer, results are also given for other targets.

DDO and RBDO are solved with the same elitist genetic algorithm (population 40, at most 60 generations, binary tournament with Deb's feasibility rules (Deb, 2000), uniform crossover with probability 0.9, per-gene mutation probability $1.5/n_v$ of which 70% are steps of ± 1–3 catalogue positions and 30% random resets, $(\mu+\lambda)$ replacement, and termination after 15 generations without improvement). The best GA design is polished by a discrete local search that accepts the first feasible lighter neighbour among single downsizing moves and down/up-sizing swaps of two variables, repeated until no improvement is found. The bi-objective problem is solved with NSGA-II (Deb et al., 2002) with constrained domination (population 48, 60 generations); all feasible designs visited are archived and the non-dominated archive

is reported. The SAA samples, the GA operators and the verification samples are all seeded, so every number in this paper is reproducible from the released code.

3. Verification of the structural and reliability models

The conclusions of an optimisation study are only as credible as the models it optimises. Three verification levels were therefore applied before any design comparison was made.

3.1. Closed-form benchmarks

Table 3 compares PyNite with closed-form solutions for a cantilever, a fixed–fixed beam under uniform load, and a fixed-base portal frame under lateral load. The portal solution follows from the slope-deflection method for inextensible members, with $k = (I_b/L)/(I_c/h)$,

$$\Delta = \frac{Hh^3(2+3k)}{12EI_c(1+6k)}, \qquad M_{base} = \frac{Hh(1+3k)}{2(1+6k)}, \qquad M_{top} = \frac{3kHh}{2(1+6k)}. \quad (9)$$

Agreement is at round-off level for the statically simple cases and within 10^{-5} for the portal, the residual being the axial flexibility that PyNite includes. The second-order response of a cantilever beam-column with axial load P and tip load H has the exact tip sway $\Delta = \frac{H}{Pk}(\tan kL - kL)$, $k = \sqrt{P/EI}$. PyNite's P-Δ solver with a single element is within 0.6% of the exact value even at $P = 0.6P_{cr}$, and converges to $< 2\times10^{-6}$ with eight elements (Table 3), showing that its geometric stiffness also captures the member (P-δ) effect.

3.2. Independent solver

The optimised designs were rebuilt in OpenSeesPy (McKenna et al., 2010; Zhu et al., 2018) with elastic beam-column elements, element loads for the distributed gravity and self-weight and nodal storey forces, independently of the PyNite model. Table 4 lists the largest relative difference of each response quantity, normalised by the largest absolute value of that quantity in the frame. For all five unit cases the two solvers agree to 10^{-13}–10^{-16}, i.e. to machine precision. (Gravity-load drifts of these symmetric frames are themselves of the order of round-off and are not compared.) For the P-Δ analysis of the combination $1.2D + 0.5L + 1.6W$, OpenSees used the `PDelta` coordinate transformation and a load-controlled Newton solution. PyNite's

Table 3: PyNite versus closed-form solutions ($E = 200$ GPa).

Quantity	Closed form	PyNite	Rel. error
Cantilever tip deflection (m)	0.053333	0.053333	0.0e+00
Cantilever fixed-end moment (kN m)	200	200	0.0e+00
Fixed beam end moment (kN m)	213.33	213.33	4.0e-16
Fixed beam mid-span moment (kN m)	106.67	106.67	1.1e-15
Fixed beam mid-span deflection (m)	0.010667	0.010667	3.3e-16
Portal sway (m)	0.012698	0.012699	9.3e-06
Portal column base moment (kN m)	114.29	114.29	7.2e-06
Portal column top moment (kN m)	85.714	85.715	3.6e-06
Second-order cantilever, tip sway (mm): exact vs PyNite P-Δ with n elements			
P/P_{cr}	exact	$n = 1$ / $n = 2$ / $n = 8$	err. ($n = 1$, $n = 8$)
0.2	25.972	25.964 / 25.972 / 25.972	3.3e-04, 1.0e-07
0.4	34.534	34.471 / 34.529 / 34.534	1.8e-03, 5.5e-07
0.6	51.650	51.326 / 51.627 / 51.650	6.3e-03, 1.8e-06

second-order drifts differ from OpenSees by at most 0.8%, and the B_1–B_2 amplified first-order drifts used by the surrogate by at most 1.8%, always on the conservative side (Fig. 2); column base moments behave similarly (at most 1.6%).

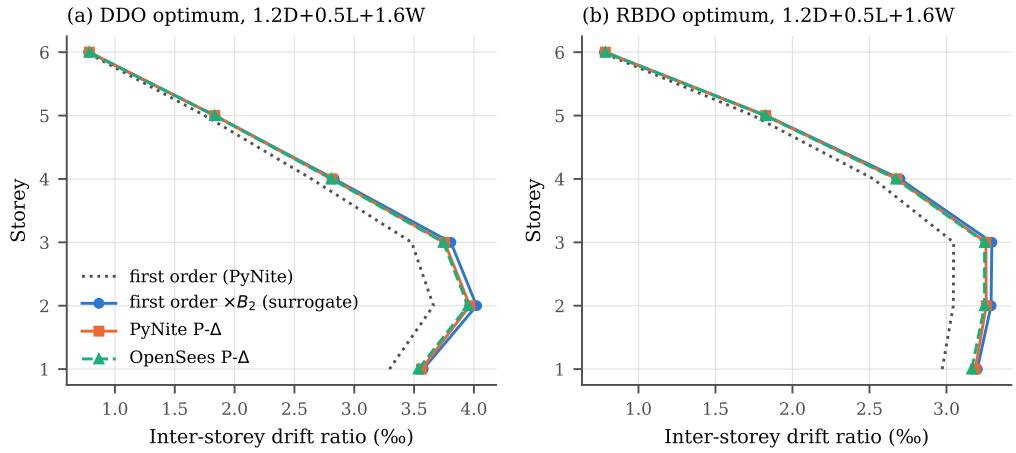


Figure 2: Inter-storey drift profiles of the base designs under $1.2D + 0.5L + 1.6W$: first-order PyNite, the B_2-amplified surrogate, PyNite P-Δ and OpenSees P-Δ.

3.3. Reliability surrogate versus full second-order re-analysis

The surrogate rests on three simplifications: second-order effects through B_1–B_2, a modulus that is uniform over the frame, and nominal section stiff-

Table 4: Maximum relative differences between PyNite and OpenSeesPy for the optimised base designs and two other frames: linear unit cases and P-Δ analysis of $1.2D + 0.5L + 1.6W$.

Design	Case	Drift	Column M	Column N	Beam end M
Base DDO	D	–	8.3e-16	2.0e-15	8.0e-16
	L	–	4.7e-16	1.6e-15	9.6e-16
	W	1.8e-14	1.1e-14	9.8e-15	1.4e-14
	ND	1.8e-14	1.3e-14	8.2e-15	1.4e-14
	NL	1.5e-14	1.0e-14	8.7e-15	1.3e-14
	P-Δ: PyNite	7.3e-03	7.4e-03		
	P-Δ: B_1–B_2	1.8e-02	1.6e-02		
Base RBDO	D	–	6.7e-16	1.4e-15	4.7e-16
	L	–	6.7e-16	1.3e-15	6.7e-16
	W	4.2e-14	4.3e-14	3.0e-14	4.2e-14
	ND	4.2e-14	4.2e-14	2.9e-14	4.1e-14
	NL	4.2e-14	4.3e-14	3.0e-14	4.2e-14
	P-Δ: PyNite	7.1e-03	6.2e-03		
	P-Δ: B_1–B_2	1.5e-02	1.5e-02		
3-storey DDO	D	–	3.1e-16	1.1e-15	4.9e-16
	L	–	7.7e-16	8.8e-16	5.5e-16
	W	6.0e-15	6.9e-15	2.8e-15	4.0e-15
	ND	6.0e-15	6.8e-15	2.9e-15	4.4e-15
	NL	6.8e-15	7.4e-15	3.5e-15	4.4e-15
	P-Δ: PyNite	5.7e-03	6.2e-03		
	P-Δ: B_1–B_2	9.4e-03	1.5e-02		
9-storey DDO	D	–	9.5e-16	2.0e-15	1.1e-15
	L	–	1.4e-15	2.8e-15	2.0e-15
	W	1.7e-13	1.1e-13	1.3e-13	1.4e-13
	ND	1.7e-13	1.2e-13	1.3e-13	1.4e-13
	NL	1.7e-13	1.2e-13	1.3e-13	1.4e-13
	P-Δ: PyNite	8.1e-03	7.5e-03		
	P-Δ: B_1–B_2	1.8e-02	1.5e-02		

ness (the fabrication factor enters the resistance only). To quantify their combined effect, 1000 samples of the wind scenario S_2 were analysed for the base RBDO design with a full PyNite P-Δ analysis per sample, in which every member received its own modulus factor (c.o.v. 0.03) and its fabrication factor was applied to A and I. The resulting system utilisation and maximum drift ratio are compared with the surrogate evaluated on the same samples (with the mean of the member modulus factors) in Fig. 3. A second set of 1000 samples with the wind doubled was added as a stress test that produces failure probabilities large enough to be compared directly.

The surrogate overestimates the system utilisation by 0.9% on average (standard deviation of the difference 1.0%, correlation 0.997) under W_{50} and by 1.5% under $2W_{50}$; the corresponding failure fractions in the stress test are 2.1% (surrogate) and 1.7% (full analysis). The drift ratio is unbiased (-0.6% mean difference, 2.0% standard deviation) and the drift exceedance fractions are 7.2% and 7.1%. The surrogate is thus slightly conservative for the ULS and accurate for the drift SLS, while being two to three orders of magnitude cheaper than a P-Δ re-analysis per sample.

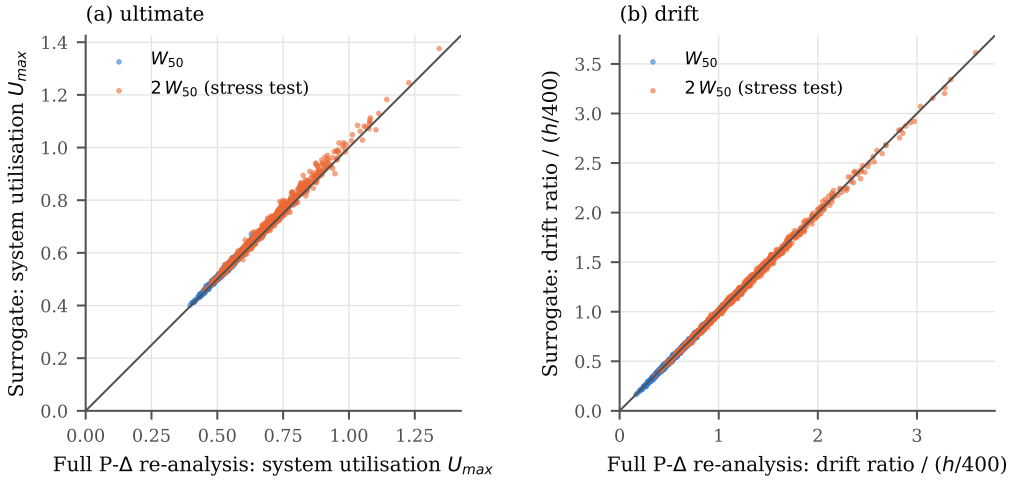


Figure 3: Surrogate versus full P-Δ re-analysis with member-by-member random stiffness and section properties (base RBDO design, 1000 samples per wind level).

3.4. Monte Carlo convergence

Fig. 4 shows the reliability indices of the base DDO and RBDO designs as functions of the sample size for five independent random streams. At

the SAA sample size ($N = 5 \times 10^4$) the scatter of $\hat{\beta}_{ULS}$ between streams is about ± 0.1, which is the resolution with which the optimiser can enforce $\beta_T = 3$; at $N = 10^6$ the streams agree within ± 0.02 (c.o.v. of $\hat{P}_f$ about 3%). The drift index converges much faster because its failure probability is larger. These numbers justify the two-stage procedure of Section 2.5: the SAA estimate guides the search and the independent 10^6-sample estimate is the one reported.

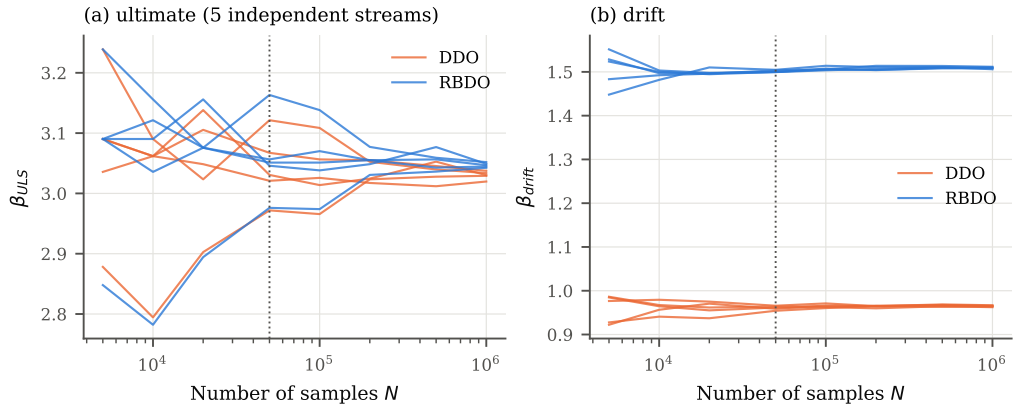


Figure 4: Convergence of the Monte Carlo reliability indices of the base designs (five independent streams each). The dotted line marks the SAA sample size used in the optimiser.

4. Results

4.1. Optimiser performance

Table 5 summarises the optimisation runs. Repeated DDO runs with different seeds converged to the same design or to designs within 0.6% in mass, and the two RBDO runs of the base frame converged to the same design, which indicates that the GA with local search reliably locates the optimum of these nine-variable problems (Fig. 5). A DDO run needs 6–12 min and an RBDO run 15–61 min on one CPU core, the reliability evaluation accounting for about 80% of the RBDO time.

4.2. Deterministic versus uncertainty-aware design of the base frame

Table 6 compares the DDO and RBDO optima of the base frame. The DDO design weighs 18.26 t; its governing LRFD ratio is 0.993 and its drift is

Table 5: Optimisation runs (best design reported; spread = range of the best masses over the seeds).

Frame	Mode	Seeds	Best mass (t)	Spread (%)	Designs evaluated	Wall time (min)
3 storeys	DDO	2	7.99	0.0	1556	2
3 storeys	RBDO	1	8.19	0.0	1390	15
Base	DDO	3	18.26	0.3	2166	8
Base	DDO-h500	1	18.79	0.0	2162	6
Base	RBDO	2	18.79	0.0	2074	44
9 storeys	DDO	2	32.99	0.0	2247	12
9 storeys	DDO-h500	1	35.48	0.0	1938	8
9 storeys	RBDO	1	35.45	0.0	1885	61
Bay 6 m	DDO	2	13.71	0.0	1937	6
Bay 6 m	DDO-h500	1	14.58	0.0	2105	6
Bay 6 m	RBDO	1	14.62	0.0	2278	57
Bay 10 m	DDO	2	25.64	0.0	1938	7
Bay 10 m	DDO-h500	1	26.04	0.0	2154	5
Bay 10 m	RBDO	1	26.07	0.0	2234	43
p_w = 0.8 kPa	DDO	2	17.79	0.0	1999	6
p_w = 0.8 kPa	RBDO	1	17.87	0.0	2169	48
p_w = 1.6 kPa	DDO	2	19.74	0.6	2153	7
p_w = 1.6 kPa	DDO-h500	1	21.99	0.0	2240	6
p_w = 1.6 kPa	RBDO	1	21.74	0.0	2105	48

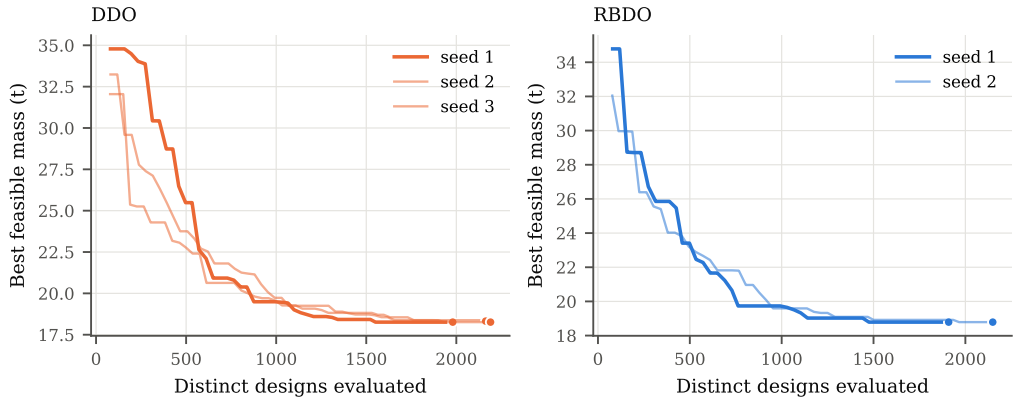


Figure 5: Convergence of the GA for the base frame (best feasible mass); dots mark the final designs after local search.

0.97 of the $h/400$ limit, so both strength and drift are active. Verified with 10^6 independent samples, its system ULS reliability index is $\beta_{ULS} = 3.03$, practically equal to the target, but its drift reliability over 50 years is only $\beta_{drift} = 0.96$ ($P_f = 0.17$): a drift check at the nominal 50-year wind leaves a one-in-six chance that the drift limit is exceeded during the reference period. The reason is that the 50-year maximum wind has a mean-to-nominal ratio of 0.78 but a c.o.v. of 0.37, so the nominal wind is exceeded with a probability of about 0.19, and serviceability checks carry no load factor.

The RBDO design ($\beta_T = 3.0$, $\beta_{SLS} = 1.5$) weighs 18.79 t (+2.9%) and achieves $\beta_{ULS} = 3.05$ and $\beta_{drift} = 1.51$, i.e. both reliability constraints are active and satisfied in the independent verification. The additional steel is placed where it buys stiffness most cheaply: the lower-storey beams are deepened (W21×50 to W24×55) and the interior columns of storeys 3–4 increased, while the remaining sections are unchanged. Its deterministic drift ratio is 0.80 of $h/400$ (equivalent to $h/500$), and its LRFD strength ratio decreases slightly to 0.977. The exceedance curves of Fig. 6 show that the two designs have virtually identical distributions of the system utilisation, whereas the drift distribution of the RBDO design is shifted to the left over its whole range.

Table 6: Optimum designs of the base frame and their deterministic and probabilistic performance (reliability indices from 10^6 independent samples; $>$ indicates no failure in the sample, i.e. a lower bound).

	DDO			RBDO		
Storeys	Ext. col.	Int. col.	Beam	Ext. col.	Int. col.	Beam
1-2	W14X61	W14X90	W21X50	W14X61	W14X90	W24X55
3-4	W14X48	W14X61	W21X44	W14X48	W14X68	W21X44
5-6	W14X48	W14X48	W21X44	W14X48	W14X48	W21X44
Steel mass (t)	18.26			18.79		
LRFD strength ratio	0.993			0.977		
Drift ratio / limit (D+0.5L+W)	0.972			0.798		
Deflection / limit (L)	0.156			0.156		
Max. B_2 (DAM)	1.186			1.151		
β_{ULS} ($N = 10^6$)	3.03			3.05		
$P_{f,ULS}$	1.24e-03			1.15e-03		
β_{drift}	0.96			1.51		
β_{defl}	>4.89			>4.89		

Fig. 7 relates the member utilisations of the deterministic check to their probabilistic counterparts. The 99.9% quantile of the member utilisation lies

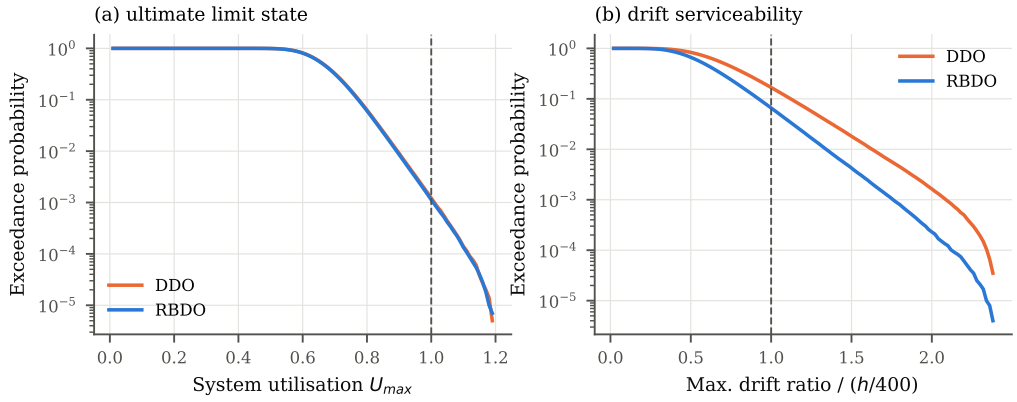


Figure 6: Exceedance probabilities of the system utilisation (ULS) and of the maximum inter-storey drift (normalised by $h/400$) of the base DDO and RBDO designs over 50 years (10^6 samples).

5–15% below the LRFD ratio for most members, which reflects the margin built into the partial factors. In 96% of the failed samples of both designs the governing member is a beam, mostly at floor levels 3–5 (79% of failures for DDO, 96% for RBDO), where the W21×44 beams of the upper storey groups are highly utilised; the remainder are exterior ground-storey columns and, for DDO, first-floor beams. The gravity scenario S_1 with the 50-year live load causes 81% (DDO) and 96% (RBDO) of the failures, which is consistent with the dominant influence of the live-load c.o.v. found in the sensitivity study. Stiffening the frame against drift (RBDO) therefore removes most wind-scenario failures without changing the gravity-governed ULS reliability.

4.3. Cost–reliability trade-off

Fig. 8 shows the non-dominated designs of the bi-objective problem (mass versus β_{ULS}, subject to $\beta_{drift}, \beta_{defl} \geq 1.5$), re-verified with 4×10^5 independent samples, together with all feasible designs visited by NSGA-II. Because drift governs the mass of this frame once $\beta_{SLS} = 1.5$ is imposed, the front is almost vertical: raising the ULS reliability from $\beta_{ULS} = 3.06$ to 3.36, 3.64 and 3.83 costs only +1.3%, +3.5% and +4.2% of steel (19.04, 19.45 and 19.57 t against 18.79 t), whereas lowering it to 2.36 would save just 0.2%. All front designs have β_{drift} between 1.51 and 1.57 (Fig. 8b). The front passes through the single-objective RBDO optimum, which is an independent check of both optimisers. The DDO optimum lies to the left of the front at a similar β_{ULS},

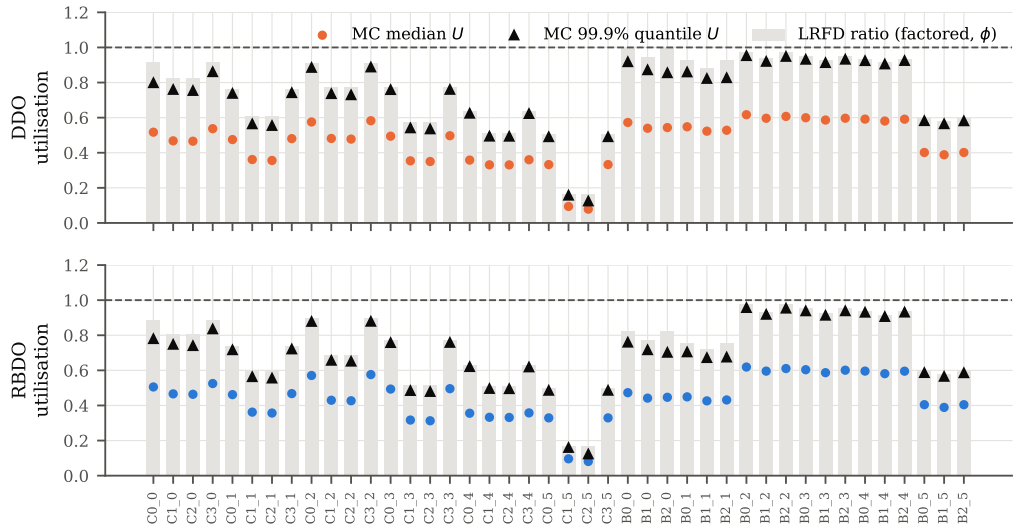


Figure 7: Member utilisations of the base designs: governing LRFD ratio (bars) and median and 99.9% quantile of the probabilistic utilisation. Member labels: Ci_s is the column on line i in storey s, Bi_s the beam in bay i at floor s (counted from 0).

but it does not belong to the feasible set because its $\beta_{drift} = 0.96$.

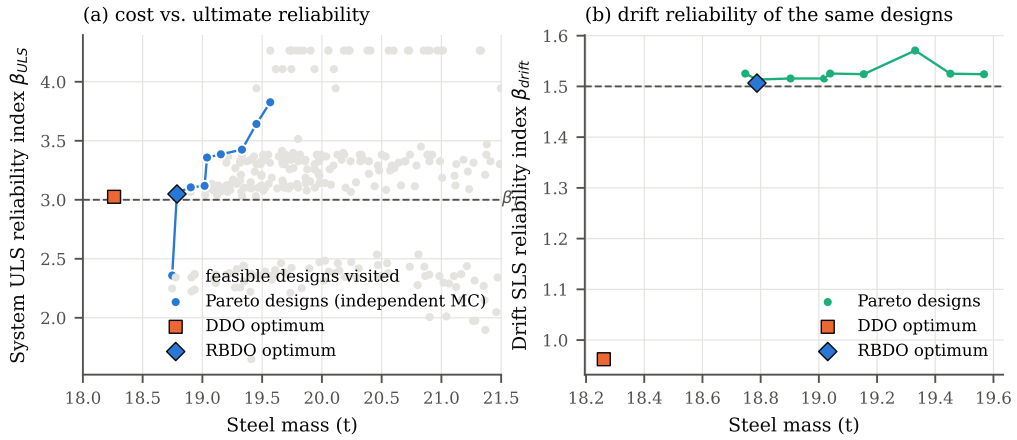


Figure 8: Cost–reliability trade-off of the base frame. (a) Non-dominated designs of the bi-objective problem (independent re-verification) with all feasible designs visited by NSGA-II (SAA estimates; values at the top are capped by the sample size); (b) drift reliability of the same non-dominated designs. The DDO and RBDO optima are shown for reference.

4.4. Effect of the reliability targets

Table 7 isolates the role of the two reliability targets for the base frame. With $\beta_T = 3.0$ and a serviceability target of 0.5, RBDO returns *exactly* the DDO design: the code-based optimum is also the reliability-based optimum when the reliability levels it implicitly provides are required. Raising β_{SLS} to 1.0 and 1.5 costs +1.0% and +2.9% of steel, and raising β_T to 3.5 at $\beta_{SLS} = 1.5$ costs +6.5% (the verified β_{ULS} of that design, 3.67, exceeds the target because of the discreteness of the catalogue). A deterministic optimisation with the drift limit tightened from $h/400$ to $h/500$ finds the same design as RBDO with $(3.0, 1.5)$, i.e. for this frame a single calibrated drift limit reproduces the reliability-based optimum.

Table 7: Base frame: optimum mass for different reliability targets and drift limits (indices verified with 10^6 independent samples; Δmass relative to DDO with $h/400$).

Formulation	β_T	β_{SLS}	Mass (t)	Δmass	β_{ULS}	β_{drift}	LRFD ratio
DDO, drift $h/400$	–	–	18.26	+0.0%	3.03	0.96	0.993
DDO, drift $h/500$	–	–	18.79	+2.9%	3.05	1.51	0.977
RBDO	3.0	0.5	18.26	+0.0%	3.03	0.96	0.993
RBDO	3.0	1.0	18.44	+1.0%	3.06	1.00	0.982
RBDO	3.0	1.5	18.79	+2.9%	3.05	1.51	0.977
RBDO	3.5	1.5	19.45	+6.5%	3.67	1.52	0.894

4.5. Parametric study

Table 8 and Fig. 9 extend the comparison to the parametric family. Three observations stand out.

First, the reliability of deterministic optima is far from uniform. Although every DDO design has a governing strength or drift check within 3% of its limit, the verified system indices range from $\beta_{ULS} = 2.88$ (10-m bays) to 3.66 (6-m bays), and the drift indices from 0.89 (high wind) to 2.54 (three storeys). Frames whose mass is governed by drift (six- and nine-storey, short bays, high wind) carry excess ULS reliability; frames governed by strength (three storeys, long bays, low wind) fall slightly below $\beta_{ULS} = 3.0$. Five of the seven DDO designs have $\beta_{drift} < 1.5$, four of them $\beta_{drift} < 1.0$.

Second, RBDO equalises the reliability: all RBDO designs satisfy $\beta_{ULS} \geq 2.99$ and $\beta_{drift} \geq 1.51$ in the independent verification (the smallest values, 2.99 and 3.00, lie within the sampling resolution of the SAA, Fig. 4). The

price relative to DDO ranges from +0.5% (low wind, where DDO already satisfies the SLS target) to +10.1% (high wind, where the drift reliability of DDO is lowest). The extra mass is thus essentially the cost of raising β_{drift} to 1.5, plus small ULS corrections in the strength-governed frames.

Third, where DDO has surplus ULS reliability, RBDO trades it for stiffness. For the 6-m bay frame, RBDO reduces β_{ULS} from 3.66 to 3.22 while raising β_{drift} from 0.95 to 1.55; its LRFD strength ratio is 1.011, i.e. the reliability-optimal design would marginally fail the deterministic strength check. Compared with deterministic designs with the drift limit tightened to $h/500$ (last three columns of Table 8), RBDO is never heavier when both designs meet the targets (equal for the base frame, −0.1% for nine storeys, −1.1% for high wind), and in two of the five frames the $h/500$ design misses one target ($\beta_{drift} = 1.494$ in the SAA for 6-m bays; $\beta_{ULS} = 2.92$ for 10-m bays). No single drift limit therefore delivers the target reliabilities in all frames, which is what the explicit reliability constraints achieve by construction.

Table 8: Parametric study: optimum mass and verified reliability indices (10^6 samples) of DDO and RBDO ($\beta_T = 3.0$, $\beta_{SLS} = 1.5$) designs, and deterministic designs with a drift limit of $h/500$ for the drift-governed frames.

	DDO ($h/400$)			RBDO				DDO ($h/500$)		
Frame	Mass (t)	β_{ULS}	β_{drift}	Mass (t)	β_{ULS}	β_{drift}	Δmass	Mass (t)	β_{ULS}	β_{drift}
3 storeys	7.99	2.93	2.54	8.19	3.00	2.71	+2.4%	–	–	–
Base	18.26	3.03	0.96	18.79	3.05	1.51	+2.9%	18.79	3.05	1.51
9 storeys	32.99	3.27	0.90	35.45	3.35	1.53	+7.5%	35.48	3.18	1.53
Bay 6 m	13.71	3.66	0.95	14.62	3.22	1.55	+6.7%	14.58	3.76	1.50
Bay 10 m	25.64	2.88	1.15	26.07	3.12	1.66	+1.7%	26.04	2.92	1.66
$p_w = 0.8$ kPa	17.79	2.92	1.78	17.87	2.99	1.98	+0.5%	–	–	–
$p_w = 1.6$ kPa	19.74	3.11	0.89	21.74	3.19	1.51	+10.1%	21.99	3.13	1.53

4.6. Sensitivity to the probabilistic model

Fig. 10 shows how the verified reliability indices of the base designs change when one assumption of Table 2 is altered (4×10^5 samples each). The system ULS index is most sensitive to the live-load model (c.o.v. 0.35 instead of 0.25: $\Delta\beta \approx -0.5$), the fabrication factor (removing it: +0.3) and the yield-stress model (mean-to-nominal 1.05 instead of 1.10, or c.o.v. 0.10 instead of 0.07: −0.2). Inter-member correlation of F_y changes β_{ULS} by less than 0.05 in either direction, and the out-of-plumbness and modulus variability are

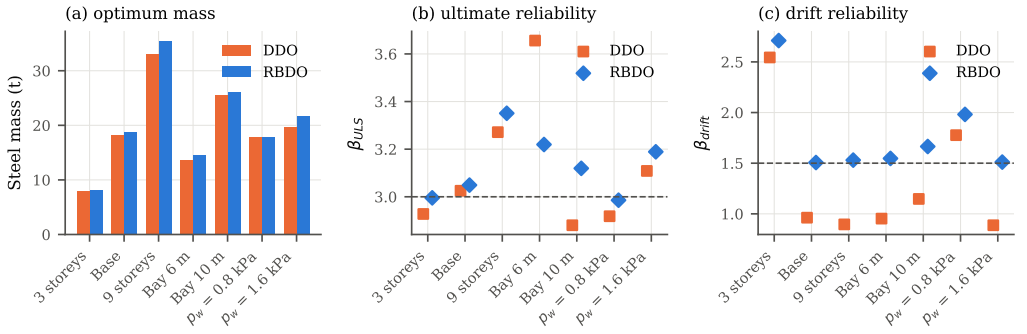


Figure 9: Parametric study: (a) optimum steel mass, (b) system ULS and (c) drift reliability indices of the DDO and RBDO designs (independent 10^6-sample verification); dashed lines mark the targets.

negligible. The drift index depends almost exclusively on the wind model: a mean-to-nominal ratio of 0.90 instead of 0.78 lowers it by 0.4, and the c.o.v. changes it by about ± 0.2. The DDO and RBDO designs react in the same way, so the conclusions of the comparison are robust to these assumptions, whereas the *absolute* indices, and hence the mass needed to meet a given target, are sensitive to the live- and wind-load statistics. This is an argument for stating the probabilistic model explicitly with any reliability-based design, as done here.

5. Discussion

Answer to the research question. For the frames studied, uncertainty-aware optimisation did *not* produce lighter designs than deterministic optimisation against the LRFD checks; it produced designs that were 0.5–10% heavier. The reason is not a weakness of RBDO but the fact that the deterministic optima are already close to the reliability-based optimum for the ULS ($\beta_{ULS} \approx 2.9$–3.7 at a target of 3.0), while the conventional drift check at the nominal 50-year wind provides only $\beta_{drift} \approx 0.9$–$1.2$ for the drift-governed frames. When the reliability levels implicit in the code are required, RBDO returns the DDO design exactly (Table 7). When a serviceability reliability of 1.5 is required, RBDO is the least-mass way of providing it: it is never heavier than a deterministic design with a tightened drift limit that meets the same targets, and it can trade surplus ULS reliability for stiffness. Economy is therefore gained *at a prescribed reliability*, not with respect to the

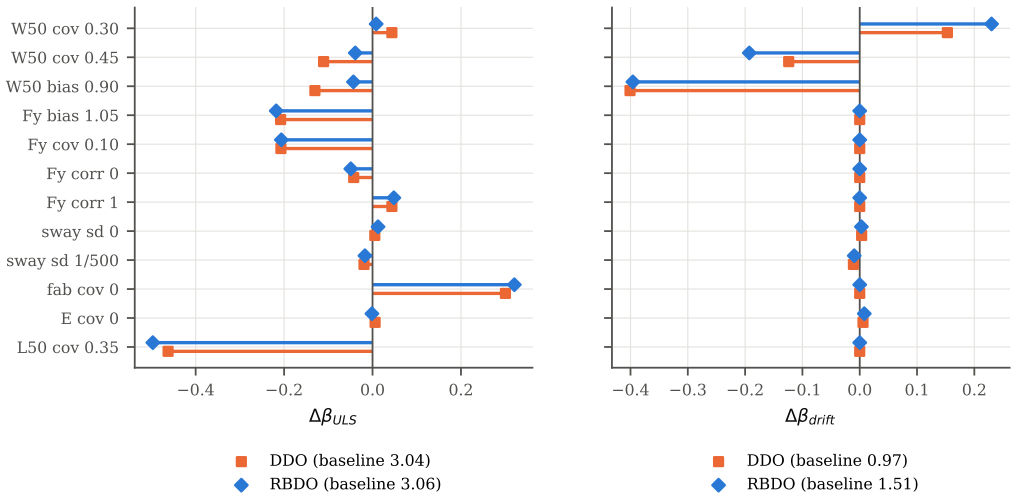


Figure 10: Change of the reliability indices of the base designs when one assumption of the probabilistic model is changed (baseline indices in the legend).

code-based design, whose reliability is unknown and non-uniform. Savings with respect to code-based designs can only be expected where the code is conservative for the system considered, which in this study was the case for the ULS of drift-governed frames (e.g. $\beta_{ULS} = 3.66$ for 6-m bays) but was masked by their serviceability deficit.

Cost of reliability. The trade-off curves are strongly asymmetric. Once a frame is drift-governed, its ULS reliability can be raised from 3.0 to almost 3.9 for about 4% more steel (Fig. 8), whereas the serviceability index is expensive to raise (2.9% of steel to raise β_{drift} from 0.96 to 1.51 in the base frame and 10.1% from 0.89 to 1.51 in the high-wind frame). Because the ULS and SLS indices of a given design are governed by different load scenarios (Fig. 7) and different random variables (Fig. 10), a single global safety measure would hide these trade-offs; reporting both indices with the mass, as done here, makes them explicit for owners and code writers.

Computational aspects. The superposition surrogate is the enabling element of the framework. Five linear PyNite analyses per design suffice to evaluate any number of load realisations and a global modulus factor exactly, and the B_1–B_2 amplification added second-order effects within 2% of two independent P-Δ solvers. Full re-analysis Monte Carlo confirmed that ignoring

member-to-member stiffness variability and the stiffness effect of fabrication tolerances makes the ULS estimate slightly conservative ($\approx 1\%$ in utilisation). With common random numbers the RBDO runs converge as robustly as the deterministic runs (identical optima for different GA seeds). A three-objective NSGA-II run (mass, β_{ULS}, β_{drift}) with a comparable budget of about 3500 designs did not approach the single-objective optima (its lightest design with $\beta_{ULS} \geq 3.0$ and $\beta_{drift} \geq 1.5$ weighed 19.50 t against 18.79 t), so for discrete frame problems of this size an ε-constraint approach with a strong single-objective optimiser, complemented by a bi-objective front, is the more economical way to map the trade-off.

Limitations. The study considers regular planar moment frames with a single wind direction, a three-variable grouping per storey group and a cost model proportional to steel mass; connection and fabrication costs, which favour fewer distinct sections, are not included. The ULS is defined by the first attainment of a member limit state in a second-order elastic analysis with the AISC interaction equations, i.e. a code-format component criterion without inelastic redistribution or model uncertainty of the interaction equations; system collapse reliabilities obtained by advanced inelastic analysis (Buonopane and Schafer, 2006; Zhang et al., 2016) would be higher. The live load on all floors and the frame-wide modulus are fully correlated, which is conservative for the former. The load statistics are the classical ones of Ellingwood et al. (1980), and the mean-to-nominal ratio of F_y is an assumption; the sensitivity study shows that the absolute indices depend on these choices by up to ± 0.5, although the comparison between DDO and RBDO does not. Finally, the drift SLS uses a deterministic limit of $h/400$; a probabilistic drift capacity of non-structural components (Galambos and Ellingwood, 1986) would refine the serviceability reliability.

6. Conclusions

A reproducible framework for reliability-constrained, multi-objective optimisation of steel moment frames was developed on the open-source PyNite library and applied to a family of planar frames with discrete W-shape sections. The main conclusions are:

1. Five linear PyNite analyses per design combined by superposition, with AISC B_1–B_2 amplification, give a reliability surrogate that agrees with

OpenSeesPy to machine precision for linear responses, within 2% for P-Δ responses, and within about 1% (conservatively) with Monte Carlo simulation using full P-Δ re-analysis and member-by-member random stiffness, at a cost that permits 5×10^4 samples per design inside a genetic algorithm.

2. Deterministic minimum-weight designs that satisfy the LRFD checks have non-uniform reliability: system ULS indices between 2.88 and 3.66 and 50-year drift reliability indices between 0.89 and 2.54 across the seven frames. Drift checked at the nominal 50-year wind corresponds to $\beta_{drift} \approx 1$ in drift-governed frames.
3. With targets of $\beta_T = 3.0$ (ULS) and $\beta_{SLS} = 1.5$, RBDO designs met both targets in independent 10^6-sample verification and were 0.5–10.1% heavier than the deterministic optima. When only the reliability levels implicit in the code were required, RBDO reproduced the deterministic design exactly. Uncertainty-aware optimisation therefore did not yield more economical designs than code-based optimisation for these frames; it yielded the least-mass designs *at an explicitly prescribed reliability*, and was never heavier than a deterministic design with a tightened drift limit meeting the same targets, a limit which in any case failed to meet them in two of five frames.
4. The cost–reliability trade-off is strongly asymmetric: in drift-governed frames the ULS reliability index can be raised from about 3.0 to 3.8 for roughly 4% more steel, while raising the drift reliability index from 0.9–1.2 to 1.5 costs 1.7–10.1%.
5. The absolute reliability indices are most sensitive to the live- and wind-load statistics and to the yield-stress and fabrication models; the relative comparison of DDO and RBDO is insensitive to them.

Future work will extend the framework to three-dimensional frames, multiple wind directions and seismic actions, a cost model including connections and fabrication, and inelastic system limit states.

Data and code availability

The framework, the input data, the scripts that produce every figure and table, and the raw results are openly available in the folder `Research/reliability_optimization` of the authors' fork of the PyNite repository.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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